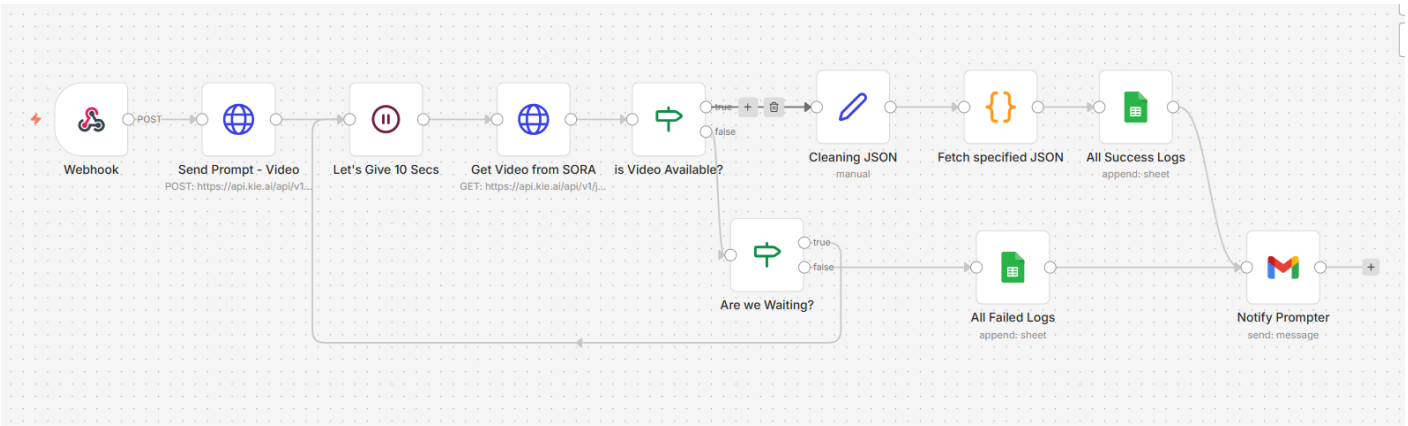

SORA 2 AI Video Generation Workflow Automation

Technical Documentation

Overview

This automated workflow facilitates AI-driven video generation using SORA 2 API. The system receives video generation prompts via webhook, submits them to SORA 2, monitors generation status, and notifies users upon completion while maintaining comprehensive logs in Google Sheets.

Workflow Architecture

System Components

- **Trigger**: Webhook (Listening endpoint)
- **Primary API**: SORA 2 Video Generation Service
- **Data Storage**: Google Sheets
- **Notification System**: Email Service
- **Orchestration Platform**: n8n/Make

Detailed Workflow Process

1. Webhook Trigger

Node Type: Webhook (Listening Mode)

Purpose: Receives incoming video generation requests

Expected Payload:

```
{
  "prompt": "AI video generation prompt text",
  "email": "user@example.com",
  "taskId": "unique_identifier"
}
```

Configuration:

- Method: POST
- Authentication: [Specify if API key/token required]
- Content-Type: application/json

---

2. Video Generation Request (HTTP POST)

Node Type: HTTP Request

Purpose: Submit video prompt to SORA 2 API for processing

Configuration:

- **Method**: POST
- **URL**: `[SORA 2 API Endpoint]/generate`
- **Authentication**: Bearer Token / API Key

Headers:
Content-Type: application/jsonAuthorization: Bearer [API_TOKEN]

-

Request Body:

```
curl -X POST "https://api.sora2.example.com/v1/generate" \
  -H "Content-Type: application/json" \
```

```
  -H "Authorization: Bearer YOUR_API_KEY" \
  -d '{
    "prompt": "{{$json.prompt}}",
    "request_id": "{{$json.request_id}}"
  }'
```

Expected Response:

```
{
  "job_id": "video_job_12345",
  "status": "processing",
  "created_at": "2026-01-01T10:00:00Z"
}
```

Data Extraction:

- Store `job_id` for subsequent status checks
- Capture initial `status`

---

3. Wait Period

Node Type: Wait/Delay

Purpose: Allow processing time before status check

Configuration:

- **Duration**: 10 seconds
- **Type**: Fixed delay

Rationale: Prevents immediate status check before SORA 2 begins processing, reducing unnecessary API calls

---

4. Status Check (HTTP GET)

Node Type: HTTP Request

Purpose: Retrieve video generation status and URL

Configuration:

- **Method**: GET
- **URL**: `[SORA 2 API Endpoint]/status/{{job_id}}`

Headers:
Authorization: Bearer [API_TOKEN]

-

Request Example:

```
curl -X GET "https://api.sora2.example.com/v1/status/video_job_12345" \
  -H "Authorization: Bearer YOUR_API_KEY"
```

Expected Response:

```
{
  "job_id": "video_job_12345",
  "status": "completed",
  "video_url": "https://cdn.sora2.example.com/videos/12345.mp4",
  "prompt": "Original prompt text",
  "created_at": "2026-01-01T10:00:00Z",
  "completed_at": "2026-01-01T10:02:15Z"
}
```

Possible Status Values:

- `processing` - Video generation in progress
- `completed` - Video successfully generated
- `failed` - Generation encountered an error
- `queued` - Request waiting in queue

---

5. Conditional Logic (IF-ELSE)

Node Type: IF Conditional Branch

Purpose: Route workflow based on generation status

IF Condition: Success

Condition: `status === "completed"`

Success Path (proceeds to steps 6-8):

- JSON data cleaning
- Google Sheets logging
- Email notification

ELSE Condition: Still Processing/Failed

Condition: `status !== "completed"`

Alternative Paths:

1. **Loop Back**: Return to Wait (step 3) with iteration counter to prevent infinite loops
    - Maximum iterations: 10
    - Total wait time: ~100 seconds
2. **Error Handling**: If max iterations reached or status = "failed"
    - Log error to Google Sheets
    - Send failure notification email

---

6. JSON Data Cleaning

Node Type: Set/Function

Purpose: Extract and format relevant data from API response

Operations:

- Parse raw API response
- Extract key fields
- Format timestamps
- Validate URLs
- Remove unnecessary metadata

Output Structure:

```
{
  "video_url": "https://cdn.sora2.example.com/videos/12345.mp4",
  "status": "completed",
  "prompt": "Original user prompt",
  "job_id": "video_job_12345",
  "user_email": "user@example.com",
  "completed_at": "2026-01-01 10:02:15",
  "processing_time": "2 minutes 15 seconds"
}
```

---

7. Google Sheets Logging

Node Type: Google Sheets (Append Row)

Purpose: Create permanent record of video generation

Configuration:

- **Spreadsheet**: SORA 2 Video Generation Log
- **Sheet**: Video Requests
- **Operation**: Append row

Data Mapping:

Column	Data Source	Example
Timestamp	Current datetime	2026-01-01 10:02:15
Job ID	job_id	video_job_12345
User Email	user_email	user@example.com
Prompt	prompt	"A serene sunset over mountains"
Status	status	completed
Video URL	video_url	https://cdn.sora2.example.com/videos/12345.mp4
Processing Time	calculated	2m 15s

Sheet Structure:

| Timestamp | Job ID | User Email | Prompt | Status | Video URL | Processing Time |

---

8. Email Notification (Success)

Node Type: Email/Gmail Send

Purpose: Notify user of successful video generation

Configuration:

- **To**: `{{user_email}}`
- **Subject**: "Your SORA 2 Video is Ready! 🎬"
- **Email Type**: HTML

Email Template:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    body { font-family: Arial, sans-serif; line-height: 1.6; color: #333; }
    .container { max-width: 600px; margin: 0 auto; padding: 20px; }
    .header { background: #4CAF50; color: white; padding: 20px; text-align: center; }
    .content { background: #f9f9f9; padding: 20px; margin: 20px 0; }
    .video-details { background: white; padding: 15px; margin: 10px 0; border-left: 4px solid
#4CAF50; }
    .cta-button { display: inline-block; background: #4CAF50; color: white; padding: 12px 24px;
text-decoration: none; border-radius: 4px; margin: 10px 0; }
    .footer { text-align: center; color: #666; font-size: 12px; margin-top: 20px; }
  </style>
</head>
<body>
  <div class="container">
    <div class="header">
      <h1>✨ Your AI Video is Ready!</h1>
    </div>

    <div class="content">
      <p>Great news! Your SORA 2 video generation has completed successfully.</p>

      <div class="video-details">
        <h3>Video Details:</h3>
        <p><strong>Your Prompt:</strong><br>{{prompt}}</p>
        <p><strong>Status:</strong> <span style="color: #4CAF50;">✓ Completed</span></p>
        <p><strong>Job ID:</strong> {{job_id}}</p>
        <p><strong>Processing Time:</strong> {{processing_time}}</p>
      </div>

      <p style="text-align: center;">
        <a href="{{video_url}}" class="cta-button">📥 Download Your Video</a>
      </p>

      <p><strong>Direct Link:</strong><br>
      <a href="{{video_url}}">{{video_url}}</a></p>
```

```
    <p style="font-size: 14px; color: #666;">
      <em>Note: This video will be available for 30 days. Please download and save it to your
preferred location.</em>
    </p>
  </div>

  <div class="footer">
    <p>This is an automated notification from SORA 2 Video Generation System</p>
    <p>Job ID: {{job_id}} | Generated: {{completed_at}}</p>
  </div>
 </div>
</body>
</html>
```

Error Handling & Edge Cases

Scenario 1: Still Processing After Initial Check

Action: Loop back to Wait node

- Implement counter to track iterations
- Maximum 10 retry attempts (total ~100 seconds wait)
- If still processing after max retries, send "still processing" email with status check link

Scenario 2: Generation Failed

Action: Error notification path

- Log failure to Google Sheets with error details
- Send failure email to user with:
    - Error message
    - Support contact information
    - Request ID for troubleshooting

Scenario 3: API Timeout/Network Error

Action: Retry mechanism

- Implement exponential backoff (10s, 20s, 40s)
- Maximum 3 retry attempts
- Log all retry attempts

- Escalate to manual review if all retries fail

Scenario 4: Invalid Webhook Payload

Action: Validation and rejection

- Validate required fields before API submission
- Return 400 error with specific validation message
- Do not proceed with SORA 2 API call

Monitoring & Maintenance

Key Metrics to Track

1. **Success Rate**: Percentage of completed vs. failed generations
2. **Average Processing Time**: Time from submission to completion
3. **API Response Times**: Monitor SORA 2 API performance
4. **Error Frequency**: Track common failure patterns

Recommended Logs

- All webhook requests (timestamp, payload)
- All API calls (request/response)
- Conditional branch decisions
- Email delivery confirmations
- Google Sheets write operations

Maintenance Schedule

- **Daily**: Review error logs
- **Weekly**: Analyze success rates and processing times
- **Monthly**: Review and optimize workflow performance

Security Considerations

1. **API Key Protection**:

    - Store SORA 2 API keys in environment variables
    - Never expose keys in logs or email notifications

    - Rotate keys quarterly
2. **Webhook Security**:
    - Implement webhook signature verification
    - Use HTTPS only
    - Validate payload structure before processing
3. **Email Privacy**:
    - Validate email addresses before sending
    - Use BCC for multiple recipients
    - Include unsubscribe option if applicable
4. **Data Retention**:
    - Video URLs may expire after 30 days
    - Archive Google Sheets data after 90 days
    - Comply with data privacy regulations

Workflow Diagram

```
┌──────────────────────┐
│  Webhook POST        │
│  (Video Prompt)      │
└──────────┬───────────┘
           │
           ▼
┌──────────────────────┐
│  HTTP POST           │
│  Submit to           │
│  SORA 2 API          │
└──────────┬───────────┘
           │
           ▼
┌──────────────────────┐
│  Wait 10s            │
└──────────┬───────────┘
           │
           ▼
┌──────────────────────┐
│  HTTP GET            │
│  Check Status        │
└──────────┬───────────┘
           │
```

```
    ▼
  ┌─────────┐
  │ IF-ELSE │
  └─────────┘
     │
  ┌──────────────────────────┐
  │              │
  ▼              ▼
[Completed]    [Processing]
  │              │
  ▼              └──▶ Loop Back
                        (Until state == success)
 ┌──────────────┐
 │ Clean JSON   │
 └──────────────┘
     │
     ▼
 ┌──────────────┐
 │ Google       │
 │ Sheets Log   │
 └──────────────┘
     │
     ▼
 ┌──────────────┐
 │ Send Email   │
 │ (Success)    │
 └──────────────┘
```

API Documentation References

SORA 2 API Endpoints

- **Base URL**: `https://sora-2-api-here/`
- **Generate Video**: `POST /generate`
- **Check Status**: `GET /status/{taskId}`
- **Authentication**: Bearer Token

Rate Limits

- Video Generation: 10 requests/minute
- Status Checks: 60 requests/minute

Version History

Version	Date	Changes
1.0	2026-01-01	Initial workflow implementation

Support & Troubleshooting

Common Issues:

1. **Videos stuck in processing**: Check SORA 2 service status
2. **Emails not sending**: Verify SMTP configuration
3. **Google Sheets not updating**: Check API permissions

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This documentation should be reviewed and updated quarterly or when significant changes are made to the workflow.